Scaled Aircraft – Vectored Thrust (System-2)

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***Abstract*— This report develops and validates a discrete-time output-feedback controller for a scaled planar vectored-thrust aircraft operating about hover. The aircraft is represented by a six-state nonlinear PVTOL model in which total thrust regulates translational motion and differential thrust regulates pitch. The hover equilibrium is shifted so that the controller acts on thrust deviations from the constant weight-supporting load mg. The nonlinear model is linearised about level hover using Jacobian matrices and discretised using zero-order hold at $T_s = 0.01$ s. Controllability and observability tests support discrete LQR feedback and Kalman state estimation. Nonlinear simulation verifies recovery from $(10, 2)$ to $(0, 0)$, followed by transition to $(4, 2)$, under the black-box noisy measurement environment. The final controller maintains bounded pitch, stable regulator and observer poles, feasible actuator demand, and accurate reconstruction of unmeasured velocity states.**

Index Terms*— *VTOL aircraft, planar PVTOL, LQR, Kalman observer, LQG, sampled-data control, state-space modelling.

I. INTRODUCTION

VERTICAL take-off and landing (VTOL) aircraft are demanding control systems because translational motion and attitude are strongly coupled. In hover, lift is produced primarily by propulsive force rather than fixed aerodynamic surfaces. As a result, a pitch disturbance changes the direction of the thrust vector and immediately affects both horizontal and vertical acceleration. The controller must therefore regulate position while constraining pitch motion.

This coupling is central to the planar VTOL problem. The aircraft cannot directly accelerate sideways. To correct horizontal position error, it must rotate, redirect part of the thrust vector, generate lateral acceleration, and then return to level attitude. If the controller is too aggressive, the aircraft may leave the small-angle region used for linearisation. If it is too conservative, recovery becomes slow. The design problem is therefore stabilisation with pitch containment, actuator feasibility, and local model validity.

This report investigates a scaled planar vectored-thrust aircraft. The nonlinear equations of motion are formulated from Newtonian mechanics, shifted about the hover thrust condition, linearised about level hover using Jacobians, and discretised for digital control design. A discrete Linear Quadratic Regulator (LQR) is used for state-feedback regulation. Since only position and pitch are measured, a Kalman observer estimates the unmeasured velocity states from noisy black-box measurements. The resulting LQG controller is validated against the nonlinear plant using standard state-space, sampled-data, and optimal-control design principles [1–3].

II. SYSTEM MODELLING

The aircraft is modelled in the vertical plane using horizontal position x, vertical position y, pitch angle θ, and their corresponding velocity states. The physical model has six states and two inputs. The first input produces a pitching moment, while the second input produces total thrust. The system parameters and actuator limits are listed in Table 1.

Table 1: System parameters and actuator limits

Parameter	Symbol	Value
Mass	m	4 kg
Pitch inertia	J	0.0475 kg m^2
Thrust offset	r	0.25 m
Gravity	g	9.81 m s^{-2}
Damping	c	0.05 N s m^{-1}
Sample time	T_s	0.01 s
Moment-channel limit	F_1	±40 N
Vertical-thrust limit	F_2	5–1000 N

II.A Problem Formulation and Nonlinear Model

The control problem is formulated as local hover regulation and position tracking for a planar vectored-thrust aircraft. The aircraft must move from an initial displaced state to a commanded hover point while keeping pitch within a region where the hover-linearised model remains valid. The state vector is

$$x_s = \begin{bmatrix} x & \dot{x} & y & \dot{y} & \theta & \dot{\theta} \end{bmatrix}^T, \tag{1}$$

where x is horizontal position, y is vertical position, and θ is pitch angle. The measured output vector is

$$y_s = \begin{bmatrix} x & y & \theta \end{bmatrix}^T. \tag{2}$$

Thus, the velocity states $\dot{x}$, $\dot{y}$, and $\dot{\theta}$ are not directly measured and must be estimated in output feedback.

The two control inputs are the differential thrust input F_1, which generates a pitching moment, and the total thrust input F_2, which generates the translational thrust force. The aircraft is under-actuated in the horizontal direction because it has no independent lateral force input. Instead, horizontal acceleration is produced indirectly by pitching the vehicle and redirecting the thrust vector.

The nonlinear equations of motion are

$$m\ddot{x} = -F_2 \sin\theta - c\dot{x}, \tag{3}$$
$$m\ddot{y} = F_2 \cos\theta - mg - c\dot{y}, \tag{4}$$
$$J\ddot{\theta} = rF_1. \tag{5}$$

Equation (3) shows that horizontal acceleration is generated only when the thrust vector is tilted by a non-zero pitch angle. Equation (4) shows that vertical acceleration is determined by the vertical projection of thrust, gravity, and damping. Equation (5) shows that pitch acceleration is directly actuated by the moment channel.

At level hover, the aircraft is stationary and level:

$$\dot{x} = \dot{y} = \dot{\theta} = 0, \qquad \ddot{x} = \ddot{y} = \ddot{\theta} = 0, \qquad \theta = 0. \tag{6}$$

Substituting these conditions into (3)–(5) gives

$$F_{1,e} = 0, \qquad F_{2,e} = mg = 39.24 \text{ N}. \tag{7}$$

The controller is designed using shifted inputs:

$$u_1 = F_1, \qquad u_2 = F_2 - mg. \tag{8}$$

With this formulation, the equilibrium input is $u_e = [0 \ \ 0]^T$. This shift is important because the controller regulates deviations from hover, while the physical thrust command is recovered by adding mg back to the second input channel.

The nonlinear state-space model is

$$\dot{x}_s = f(x_s, u), \qquad y_s = h(x_s, u), \tag{9}$$

where

$$f(x_s,u) = \begin{bmatrix} \dot{x} \\ -\dfrac{(u_2+mg)\sin\theta + c\dot{x}}{m} \\ \dot{y} \\ \dfrac{(u_2+mg)\cos\theta - mg - c\dot{y}}{m} \\ \dot{\theta} \\ \dfrac{ru_1}{J} \end{bmatrix}, \qquad h(x_s,u) = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix}. \tag{10}$$

MATLAB verification gave $\|f(x_e, u_e)\| = 0.00 \times 10^0$, confirming that the hover trim is dynamically consistent.

II.B Linearisation and Jacobian Matrices

The nonlinear plant is linearised about the hover equilibrium (x_e, u_e). Perturbation variables are defined as

$$\delta x_s = x_s - x_e, \qquad \delta u = u - u_e, \qquad \delta y_s = y_s - y_e. \tag{11}$$

Applying a first-order Taylor expansion gives

$$\delta\dot{x}_s = A\delta x_s + B\delta u, \qquad \delta y_s = C\delta x_s + D\delta u. \tag{12}$$

This local approximation is standard for nonlinear control design about an operating point, where the linear model is valid only near the selected equilibrium [5,6]. The linearised matrices are obtained from

$$A = \left.\frac{\partial f}{\partial x_s}\right|_{x_e,u_e}, \quad B = \left.\frac{\partial f}{\partial u}\right|_{x_e,u_e}, \quad C = \left.\frac{\partial h}{\partial x_s}\right|_{x_e,u_e}, \quad D = \left.\frac{\partial h}{\partial u}\right|_{x_e,u_e}. \tag{13}$$

The dominant Jacobian term comes from the horizontal acceleration equation:

$$\left.\frac{\partial \ddot{x}}{\partial \theta}\right|_{\theta=0,u_2=0} = -\frac{mg}{m} = -g. \tag{14}$$

The vertical and pitch input Jacobians are

$$\left.\frac{\partial \ddot{y}}{\partial u_2}\right|_{\theta=0} = \frac{1}{m}, \qquad \frac{\partial \ddot{\theta}}{\partial u_1} = \frac{r}{J}. \tag{15}$$

The resulting continuous-time state matrix is

$$A = \begin{bmatrix} 0 & 1.0000 & 0 & 0 & 0 & 0 \\ 0 & -0.0125 & 0 & 0 & -9.8100 & 0 \\ 0 & 0 & 0 & 1.0000 & 0 & 0 \\ 0 & 0 & 0 & -0.0125 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1.0000 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \tag{16}$$

The input and output matrices are

$$B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0.2500 \\ 0 & 0 \\ 5.2632 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}. \tag{17}$$

The matrix structure has a clear physical interpretation. The $A_{2,5} = -9.81$ term means that pitch perturbation generates horizontal acceleration. The $B_{6,1} = 5.2632$ term means that the moment input directly controls angular acceleration. The $B_{4,2} = 0.25$ term means that the thrust-deviation input directly controls vertical acceleration. Hence, altitude is directly actuated, while horizontal motion is controlled indirectly through pitch.

The open-loop continuous-time eigenvalues are 0, -0.0125, 0, -0.0125, 0, and 0. The zero eigenvalues arise from translational and rotational integrators, while the small negative eigenvalues arise from translational damping. The system is therefore marginally stable and requires feedback for recovery from displaced hover states.

II.C Discrete-Time Model and Verification

The continuous-time linear model is discretised using zero-order hold, the standard sampled-data representation when a digital controller updates the input once per sample interval [2,7]:

$$\delta x_s[k+1] = G\delta x_s[k] + H\delta u[k], \qquad \delta y_s[k] = C\delta x_s[k] + D\delta u[k]. \tag{18}$$

The selected sampling time is $T_s = 0.01$ s, corresponding to a 100 Hz update rate. The reference command is shaped by a critically damped prefilter:

$$\omega_n = 0.626 \text{ rad/s}, \qquad \zeta = 1. \tag{19}$$

Because lateral acceleration near hover is approximately $\ddot{x} \approx -g\theta$, step-command acceleration translates directly into pitch demand. The prefilter reduces command-induced pitch excursions and helps keep the response within the local hover-linearisation region.

Table 2: MATLAB verification results

Check	Required	Result
Hover residual	0	0.00×10^0
Controllability rank	6	6
Observability rank	6	6
Max. plant pole	< 1	0.9963
Max. observer pole	< 1	0.9687

The controllability rank confirms that the two-input system can influence all six states through the coupled dynamics. The observability rank confirms that measurements of x, y, and θ are sufficient to reconstruct the unmeasured velocity states. These checks justify the use of discrete LQR feedback and a Kalman observer.

III. CONTROLLER AND OBSERVER DESIGN

The controller is designed in discrete time. Since only x, y, and θ are measured, the final architecture combines an LQR regulator with a Kalman observer.

III.A LQR Tuning Rationale

The regulator is designed using discrete-time LQR, which selects the feedback gain by minimising a quadratic state-and-input cost rather than assigning poles directly [3,8]. The LQR law is

$$\delta u[k] = -K\delta x_s[k], \tag{20}$$

where K minimises

$$J = \sum_{k=0}^{\infty} \left(\delta x_s[k]^T Q \delta x_s[k] + \delta u[k]^T R \delta u[k]\right). \tag{21}$$

The selected weights are

$$Q = \mathrm{diag}(0.25,\ 0.25,\ 4,\ 0.25,\ 400,\ 4), \tag{22}$$

$$R = 10\,\mathrm{diag}\left(\frac{1}{2^2},\ \frac{1}{5^2}\right). \tag{23}$$

The weighting is asymmetric because the control channels have different physical roles. The vertical channel is directly actuated by u_2, while lateral recovery requires pitch motion. The large pitch weight protects the hover-linearisation assumption, while the input penalty avoids excessive moment-channel action.

The resulting LQR gain is

$$K = \begin{bmatrix} -0.2948 & -0.9346 & 0 & 0 & 13.5969 & 2.5574 \\ 0 & 0 & 3.1424 & 5.0255 & 0 & 0 \end{bmatrix}. \tag{24}$$

The first row acts on x, $\dot{x}$, θ, and $\dot{\theta}$, confirming that lateral regulation is attitude-mediated. The second row acts on y and $\dot{y}$, matching the direct vertical-thrust channel.

III.B LQR Input-Penalty Sensitivity

An R-scaling sensitivity study was performed while holding Q fixed:

$$R_\rho = \rho R. \tag{25}$$

Smaller ρ permits more aggressive control action. Larger ρ produces a softer controller with lower input demand.

Table 3: LQR input-penalty sensitivity

ρ	max $\lvert K\rvert$	max $\lvert u_1\rvert$	max $\lvert\theta\rvert$	Err.
0.25	25.430	0.5862	0.2086	0.0449
0.50	18.617	0.3770	0.2063	0.0456
1.00	13.597	0.2285	0.2039	0.0465
2.00	9.928	0.1254	0.2019	0.0478
5.00	6.566	0.0695	0.2003	0.0500
10.00	4.819	0.0570	0.2000	0.0541

The selected value $\rho = 1$ is a defensible compromise. Lower values marginally reduce residual error but significantly increase gain magnitude and u_1 demand. Higher values reduce actuator demand but increase residual tracking error. Across all cases, pitch remains below the selected 0.35 rad local-validity threshold.

III.C Full-State Validation

The first validation stage applies full-state LQR feedback to the nonlinear plant. The aircraft initially moves from $(10, 2)$ to $(0, 0)$. At approximately $t = 15$ s, a second reference is applied and the aircraft moves toward $(4, 2)$.

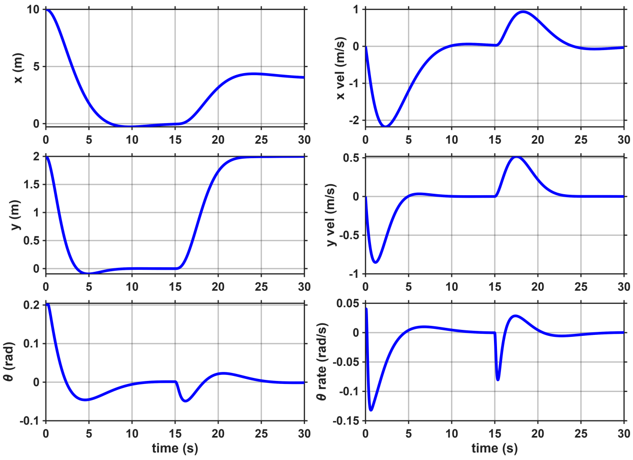

Figure 1: Stage 1 full-state LQR state response on the nonlinear plant. The aircraft regulates from $(10, 2)$ toward the origin, then responds to the second commanded hover point near $(4, 2)$.

III.D Linearisation Consistency

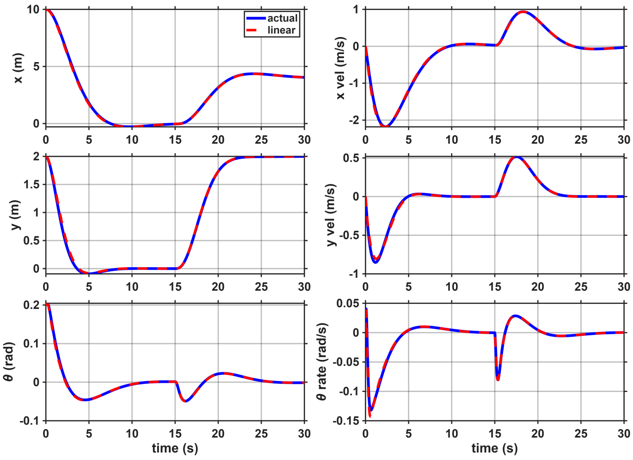

Figure 2: Linearised-model state response compared with the nonlinear plant. The close agreement confirms that the nominal response remains inside the local hover-linearisation region.

The linear and nonlinear responses remain closely aligned during the nominal manoeuvre. This confirms that the pitch motion remains sufficiently small for the hover-linearised model to represent the nonlinear plant.

III.E Observer Design and Noisy Measurement Rejection

A Kalman observer is used to estimate the full state from noisy measured outputs, following the minimum-variance state-estimation framework introduced by Kalman [4]. The controller does not receive the full state vector directly. Instead, the black-box plant model supplies measured position and attitude channels, which include measurement noise before being passed to the observer:

$$y_m[k] = Cx_s[k] + v[k]. \tag{26}$$

The observer is implemented as

$$\hat{x}_s[k+1] = G\hat{x}_s[k] + H\delta u[k] + L\left(y_m[k] - C\hat{x}_s[k]\right). \tag{27}$$

The innovation term corrects the model prediction using the noisy measured outputs. The Kalman gain prevents the feedback loop from reacting directly to every high-frequency measurement fluctuation, while still reconstructing the unmeasured velocity states.

The covariance weights are

$$Q_n = \mathrm{diag}(0.01,\ 0.1,\ 0.01,\ 0.1,\ 0.01,\ 0.1), \qquad R_n = 0.1 I_3. \tag{28}$$

The larger process-noise weights on velocity states reflect that $\dot{x}$, $\dot{y}$, and $\dot{\theta}$ are not directly measured. The measurement-noise weighting in R_n reflects the noisy output channels supplied by the black-box plant model.

The observer error dynamics and output-feedback law are

$$e[k+1] = (G - LC)e[k], \qquad \delta u[k] = -K\hat{x}_s[k]. \quad (29)$$

The physical thrust input is recovered by adding hover thrust:

$$F_1[k] = u_1[k], \qquad F_2[k] = u_2[k] + mg. \quad (30)$$

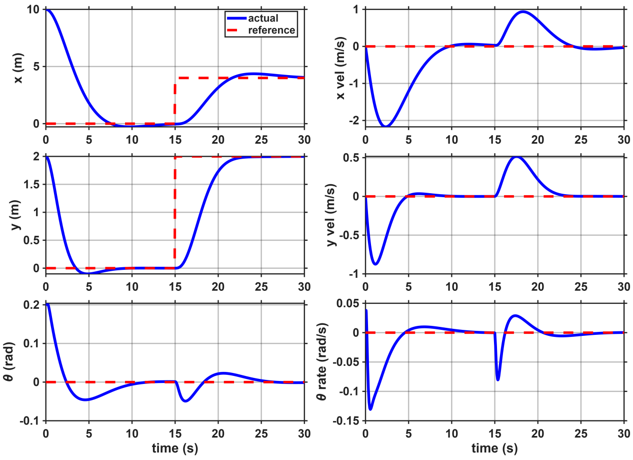

Figure 3: Stage 2 observer-based output-feedback state response under noisy measured outputs. Only x, y, and θ are measured; the velocity states are reconstructed by the Kalman observer.

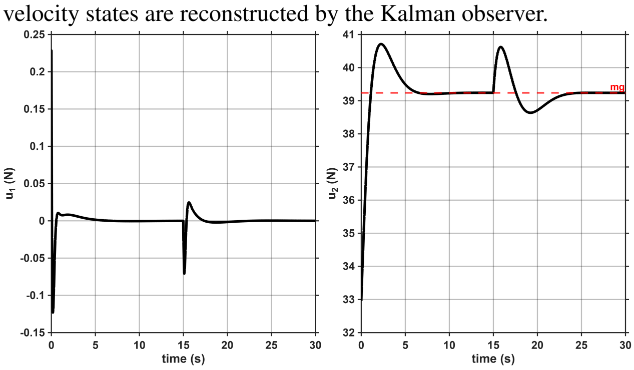

Figure 4: Stage 2 output-feedback control inputs under the black-box noisy measurement environment. The thrust channel remains close to the hover load, while the moment-channel input remains bounded during both manoeuvre segments.

By the separation principle, the regulator poles are governed by $G - HK$, while the observer poles are governed by $G - LC$. MATLAB verification gave

$$\max|\lambda(G-HK)| = 0.9963, \qquad \max|\lambda(G-LC)| = 0.9687. \quad (31)$$

Both values are inside the unit circle, so the observer-based controller is stable in discrete time.

III.F Discrete-Time Stability

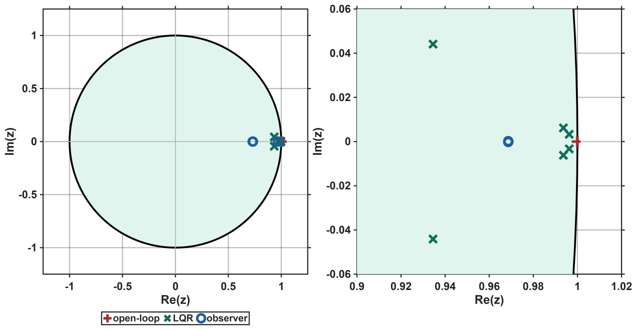

Figure 5: Discrete-time pole map comparing open-loop poles, closed-loop LQR poles, and observer poles.

The pole map shows open-loop poles clustered near $z = 1$, consistent with integrator-like hover dynamics. The LQR and observer poles lie inside the unit circle, confirming discrete-time closed-loop stability.

IV. SIMULATION RESULTS AND PERFORMANCE ANALYSIS

The nonlinear simulation was run for 30 s from

$$x_s(0) = \begin{bmatrix}10 & 0 & 2 & 0 & 0.2 & 0\end{bmatrix}^T. \quad (32)$$

The command sequence tests both recovery and tracking:

$$(10, 2) \rightarrow (0, 0) \rightarrow (4, 2). \quad (33)$$

The final position error is 0.0465 m. Because the controller uses set-point shifting without a reference feedforward gain K_r or integral action, a small steady-state offset is expected for non-zero references. This is a design limitation, not an instability. The maximum pitch magnitude is

$$\max|\theta| = 0.2039 \text{ rad} = 11.7^\circ, \quad (34)$$

which remains below the selected small-angle threshold of 0.35 rad.

IV.A Observer Estimation Performance

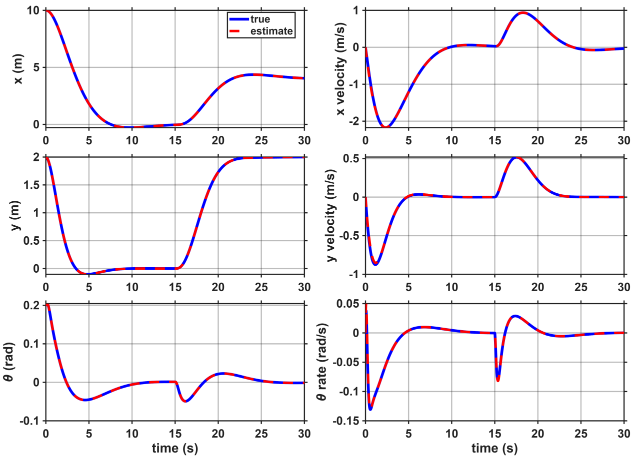

Figure 6: Kalman observer state-estimation performance under noisy measured outputs.

The observer accurately reconstructs both measured and unmeasured states despite the noisy measured outputs supplied by the black-box plant model. The estimation errors are summarised in Table 4. These results indicate successful noise rejection rather than a noise-free simulation.

Table 4: Kalman observer estimation error under noisy measured outputs

State	RMS	Max.
x (m)	0.00029	0.00201
$\dot{x}$ (m/s)	0.00860	0.05960
y (m)	0.00027	0.00155
$\dot{y}$ (m/s)	0.00787	0.04582
θ (rad)	0.00000	0.00001
$\dot{\theta}$ (rad/s)	0.00002	0.00013

The largest estimation errors occur in the velocity states, which is expected because the velocity states are reconstructed rather than measured directly. The noisy output-feedback result achieved a final position error of 0.0465 m with $\max|\theta| = 0.2039$ rad, showing that the observer reconstructs the states without allowing measurement noise to destabilise the LQR loop.

IV.B Flight-Path Interpretation

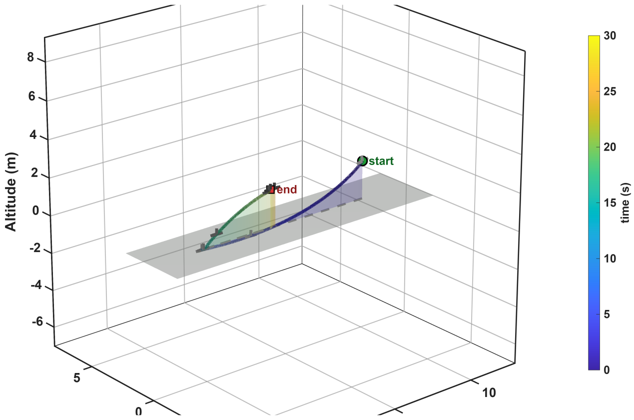

Figure 7: Three-dimensional visualisation of the simulated recovery trajectory.

The flight-path visualisation shows the geometric recovery path. The aircraft first transitions toward the origin, then moves toward the commanded hover point at $(4, 2)$. The trajectory reflects coordinated altitude regulation and pitch-mediated lateral correction.

IV.C Linearisation Validity

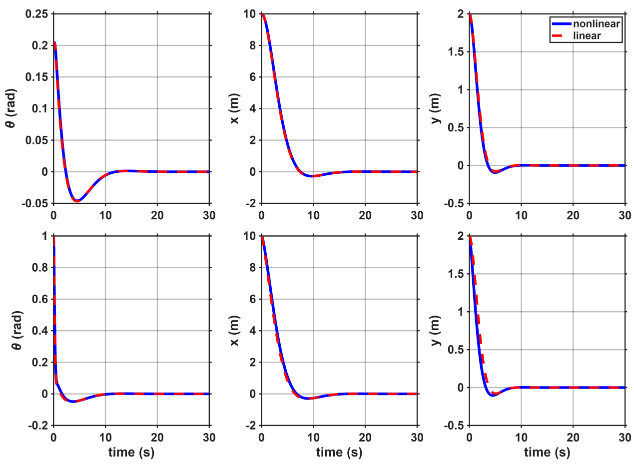

Figure 8: Linearisation-validity comparison for small and large initial pitch angles.

For the nominal initial pitch $\theta_0 = 0.2$ rad, the maximum pitch response is approximately 0.2045 rad, which remains below the 0.35 rad validity threshold. Initial pitch angles of 0.6 rad and 1.0 rad exceed the local hover-linearisation region. Those cases may still recover in nonlinear simulation, but they should not be treated as validated operation of the linearised controller.

IV.D Operating Envelope

An operating-envelope sweep was performed over

$$x_0 \in \{2,\ 5,\ 10,\ 15,\ 20,\ 25,\ 30\}\ \mathrm{m}, \tag{35}$$

$$\theta_0 \in \{0,\ 0.1,\ 0.2,\ 0.35,\ 0.5,\ 0.75,\ 1.0\}\ \mathrm{rad}. \tag{36}$$

Each case was classified according to whether recovery occurred inside or outside the selected pitch-validity region.

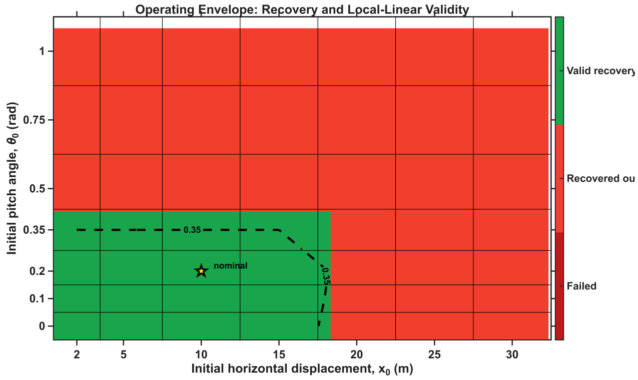

Figure 9: Operating-envelope sweep. Green indicates recovery inside the local linearisation-validity region; red indicates recovery outside the $|\theta| < 0.35$ rad threshold.

The nominal case $(x_0, \theta_0) = (10\ \mathrm{m}, 0.2\ \mathrm{rad})$ lies inside the valid recovery region. Valid recovery is maintained up to approximately $x_0 = 15$ m for small initial pitch angles. For larger offsets or $\theta_0 \geq 0.35$ rad, the aircraft may recover, but the response exceeds the local pitch-validity threshold. The controller is therefore best interpreted as a local hover regulator rather than a global nonlinear flight controller.

IV.E Sampling-Rate Sensitivity

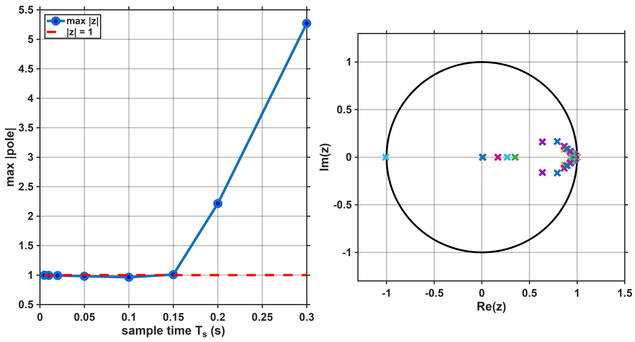

Figure 10: Sampling-rate sensitivity of the discrete-time controller.

The sampling-rate sweep shows that the controller remains stable up to $T_s = 0.10$ s, but becomes unstable at $T_s = 0.15$ s. The selected value $T_s = 0.01$ s is therefore conservative.

Table 5: Sampling-rate sensitivity

T_s	Rate	$\max \lvert z \rvert$	Status
0.005 s	200 Hz	0.9981	Stable
0.010 s	100 Hz	0.9963	Stable
0.020 s	50 Hz	0.9926	Stable
0.050 s	20 Hz	0.9816	Stable
0.100 s	10 Hz	0.9636	Stable
0.150 s	7 Hz	1.0090	Unstable
0.200 s	5 Hz	2.2134	Unstable
0.300 s	3 Hz	5.2719	Unstable

IV.F Performance Summary

Table 6: Closed-loop performance summary

Metric	Target	Achieved
Controllability rank	6	6
Observability rank	6	6
Max. plant pole	< 1	0.9963
Max. observer pole	< 1	0.9687
Final position error	Small	0.0465 m
Max. pitch angle	< 0.35 rad	0.2039
Max. u_1 effort	Bounded	0.23 N
Linear validity	Within limit	Satisfied
Actuator demand	No saturation	Feasible
Noisy output feedback	Stable	Satisfied

The nominal actuator demand is modest. The maximum moment-channel command is 0.23 N, far below the assumed ± 40 N limit. The thrust channel remains centred about the hover load $mg = 39.24$ N, well above the lower thrust bound of 5 N. The response is therefore not achieved by exploiting actuator saturation or excessive thrust authority.

V. DISCUSSION

The dominant plant feature is the $-g\theta$ term in the A matrix. It shows that horizontal motion is generated through attitude variation rather than direct horizontal forcing. The B matrix confirms the same structure: u_1 drives pitch acceleration through $B_{6,1} = 5.2632$, while u_2 drives vertical acceleration through $B_{4,2} = 0.25$. The resulting LQR gain is physically consistent: the first row of K couples x, $\dot{x}$, θ, and $\dot{\theta}$, while the second row regulates y and $\dot{y}$.

The R-scaling study shows that reducing the input penalty increases gain magnitude and moment-channel effort, but only marginally improves residual tracking error. Increasing the input penalty reduces actuator demand but worsens final accuracy. The selected value $\rho = 1$ therefore preserves actuator margin, limits pitch motion, and avoids unnecessarily aggressive feedback.

The observer validation should be interpreted as a noisy-output-feedback test rather than a full-state clean measurement test. The controller does not receive $\dot{x}$, $\dot{y}$, or $\dot{\theta}$, and the measured x, y, and θ signals are affected by the black-box sensor-noise model. The small estimation errors therefore show that the Kalman filter suppresses measurement noise while reconstructing the unmeasured states. This behaviour is consistent with the LQG separation principle, where the regulator and estimator may be designed independently for the linearised system while the combined closed-loop poles are obtained from the controller and observer eigenvalues [3, 8].

The operating-envelope sweep gives the clearest limit on the design. The controller recovers for all swept cases, but many recoveries occur outside the selected pitch-validity threshold. A nonlinear trajectory may return to hover even after leaving the region in which the linear model is accurate. Such cases demonstrate nonlinear recovery, but not validated local-linear operation. The green region in Fig. 9 is therefore the validated local-linear envelope.

The sampling-rate study confirms that the design must be interpreted as a sampled-data controller. At $T_s = 0.01$ s, both controller and observer poles remain inside the unit circle. When T_s is increased to 0.15 s, the closed-loop system becomes unstable. The sampling time is therefore part of the controller design, not an arbitrary implementation detail.

The main limitation is the absence of integral action or explicit reference feedforward. The residual error of 0.0465 m is expected for a non-zero reference under set-point shifting. This is not a stability failure; it follows from using a regulator architecture rather than a full servo controller. A zero steady-state tracking requirement would require a reference gain K_r, output integral action, or constrained model predictive control.

VI. CONCLUSION

This report developed a discrete LQG controller for a scaled planar vectored-thrust aircraft. The nonlinear model was formulated from Newtonian mechanics, shifted about the hover thrust condition, linearised about level hover using Jacobians, and discretised for digital control design. The resulting six-state model was controllable and observable.

The A, B, C, D, and K matrices were included to make the implementation explicit. Nonlinear simulations showed that full-state feedback stabilises the aircraft, that the linear and nonlinear responses remain aligned in the nominal operating region, and that observer-based output feedback retains the essential full-state-feedback behaviour under the black-box noisy measurement environment. The aircraft follows the two-segment command sequence from $(10, 2)$ to $(0, 0)$, then to $(4, 2)$, while maintaining bounded pitch motion and feasible actuator demand.

The R-sensitivity study, noisy-measurement observer validation, sampling-rate sweep, and operating-envelope analysis show that the selected controller is not merely stabilising, but occupies a defensible compromise between control effort, pitch validity, sampled-data stability, estimator performance, and local recovery capability. Future work should include integral action or reference feedforward for zero steady-state tracking error, explicit actuator saturation handling, and robustness analysis under modelling uncertainty.

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